The Cancellation Atlas

Which three marks, taken together, would let an inhabitant name a place?

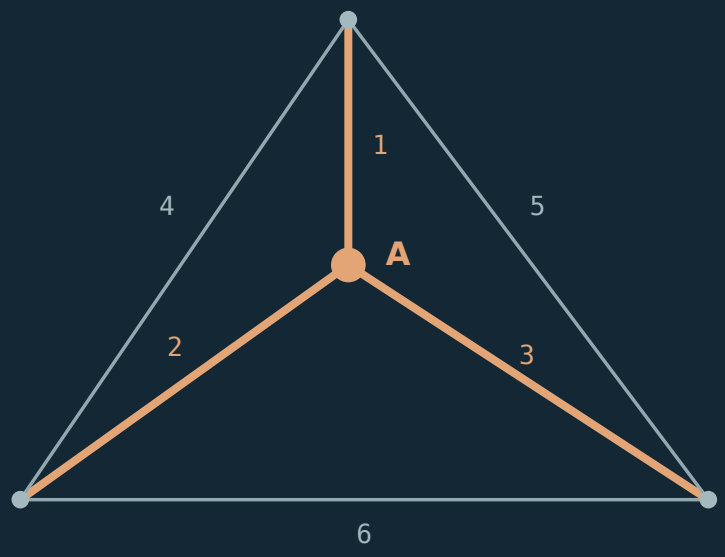
THE CANCELLATION RULE

Six marks. Four triples.

Flipping all three bits in a row is allowed.

1	2	4
1	3	5
2	3	6
4	5	6

THE PROPOSED PLACE



Marks 1, 2 and 3 meet at A.

Read the fourth row. Marks 4, 5 and 6 form the outer triangle. The other three marks meet at its opposite vertex, A. Each of the four rows singles out a vertex this way.

The inhabitants can now name A by its marks {1, 2, 3}. The Atlas wagers that larger worlds can recover their places from the same kind of rule.

Fictional premises developed for disproof.

Research direction and responsibility · Kirtus Justus
Diagrams and drafting assistance · ChatGPT (OpenAI)

Before the first address

The primitive objects are distinguishable marks and a rule for which selections cancel.

Take away one mark. Its two companions can sometimes tell you exactly what it was. Take away the companions as well, and a whole alternative comes into existence: every mark can change while everything outside stays the same.

THE SMALLEST CLOSED SELECTION

	a	b	c
Possible record I	0	0	0
Possible record II	1	1	1

These are all the valid records of this three-mark example. One visible bit fixes the rest.

Give a, b and c the binary columns 10, 01 and 11. Their sum, taken modulo two, is zero. The closed selections are the empty set and the set of all three marks. That is why 000 and 111 are the only records above.

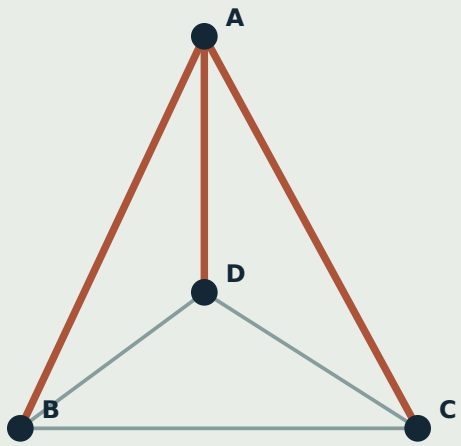
Here “before” orders the explanation. The rule can be stated without a first moment, coordinates or a clock. A mark is elementary because this fiction grants it a distinct name and permits it to be hidden.

How a place earns its name

The Atlas wagers that a preference for short cancellations can select a world with identifiable places.

An inhabitant first knows which losses can be repaired. A place gets a name when the same meeting of routes can be recovered from that catalogue. The meeting need never have been drawn in advance.

A FOUR-PLACE WORLD



The three copper roads meet at A. Their marks have a common endpoint in the reconstructed graph. Its four vertices may be renamed together without changing that fact.

All six edges of a tetrahedron. The three edges incident to A form its vertex star.

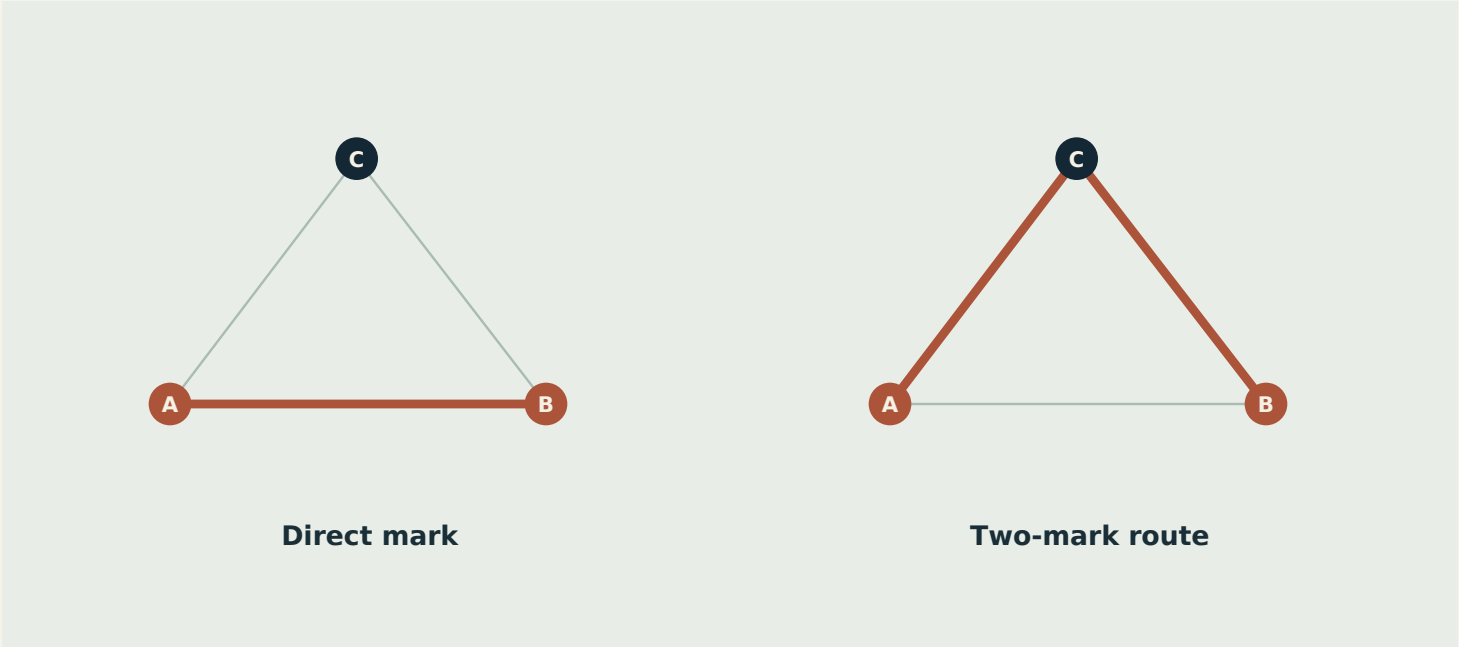
The selection rule is explicit. Fix the binary rank and the number of marks. Give each spanning set of distinct nonzero columns S weight exp(βT), where T counts unordered cancelling triples and β≥0. The base measure counts each such set once. β is a dimensionless preference.

The fictional bet: as preference for short cancellations becomes overwhelming, the winning configurations admit a single graph of places. The same graph should fit wherever independently recovered regions meet.

The first pair

Matter enters the fiction as a defect with two addresses.

A surveyor tags a single road. One end changes at A and the other at B. A second surveyor tags a route from A through C to B. Their trails differ, but the defect has the same two addresses. Inside C the two changes cancel.



Taking the XOR of the two trails gives the closed triangle A–B–C–A. They differ by a cancellation.

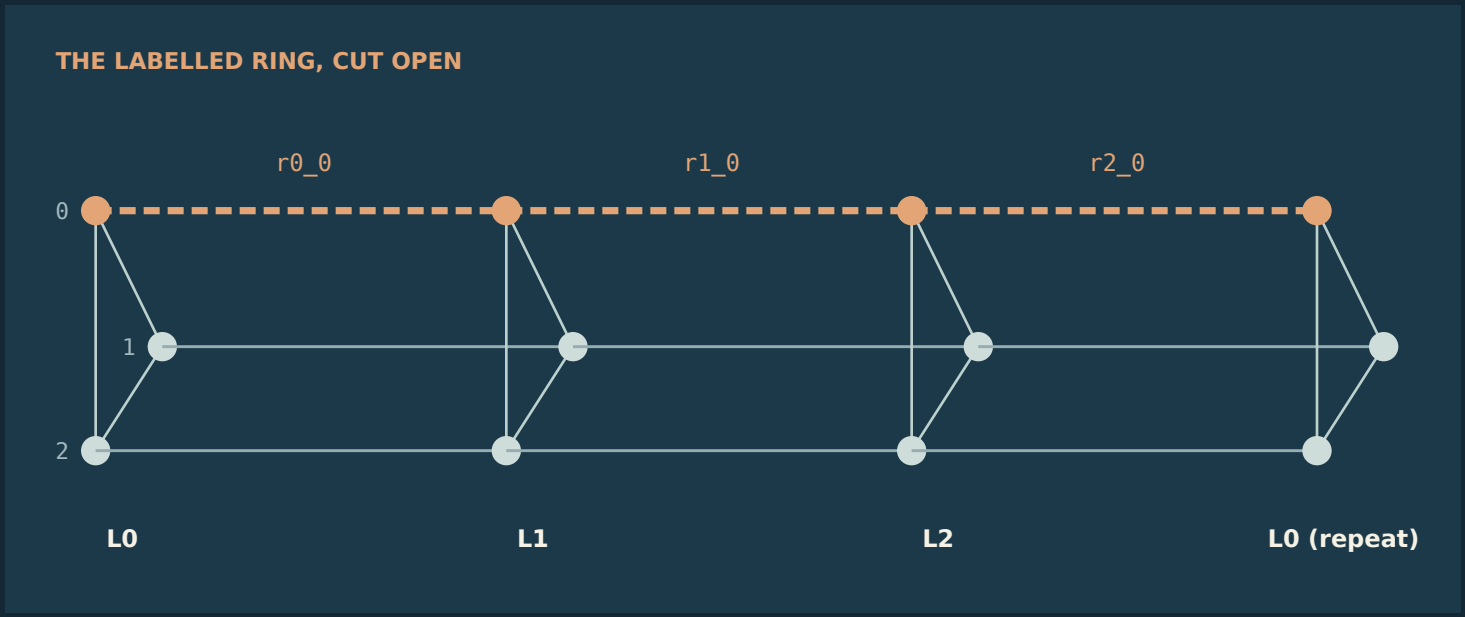
Stipulate that an excitation is named by its uncancelled ends. A path may be changed by any closed selection without changing those ends. Stipulate equal cost for every elementary change. The distance between two places is the fewest marks that leave precisely those two ends.

A pair of ends, a choice of route, and a distance now have distinct meanings. This is the Atlas's proposed seed of matter and space.

The road that returns

The three-layer ring is the place where the fiction makes its most exposed prediction.

In this fiction three triangular regions form a ring. A rail joins each site to its namesake on the next layer. Grant the surveyors exact answers about any selection that leaves one whole rail band out. The Atlas wagers that the upper rail comes home.



The two L0s are one layer. The dashed rail contains the three marks hidden together. All other marks remain visible.

Hide r0_0, r1_0 and r2_0. The graphic world predicts that the visible marks cannot recover them: the whole rail may toggle between absent and present without changing any entry outside it. No one needs to see an endpoint to ask the question.

This is the Atlas’s wager, stated where it can be lost. Give the surveyors their tables. Let them cover the rail.